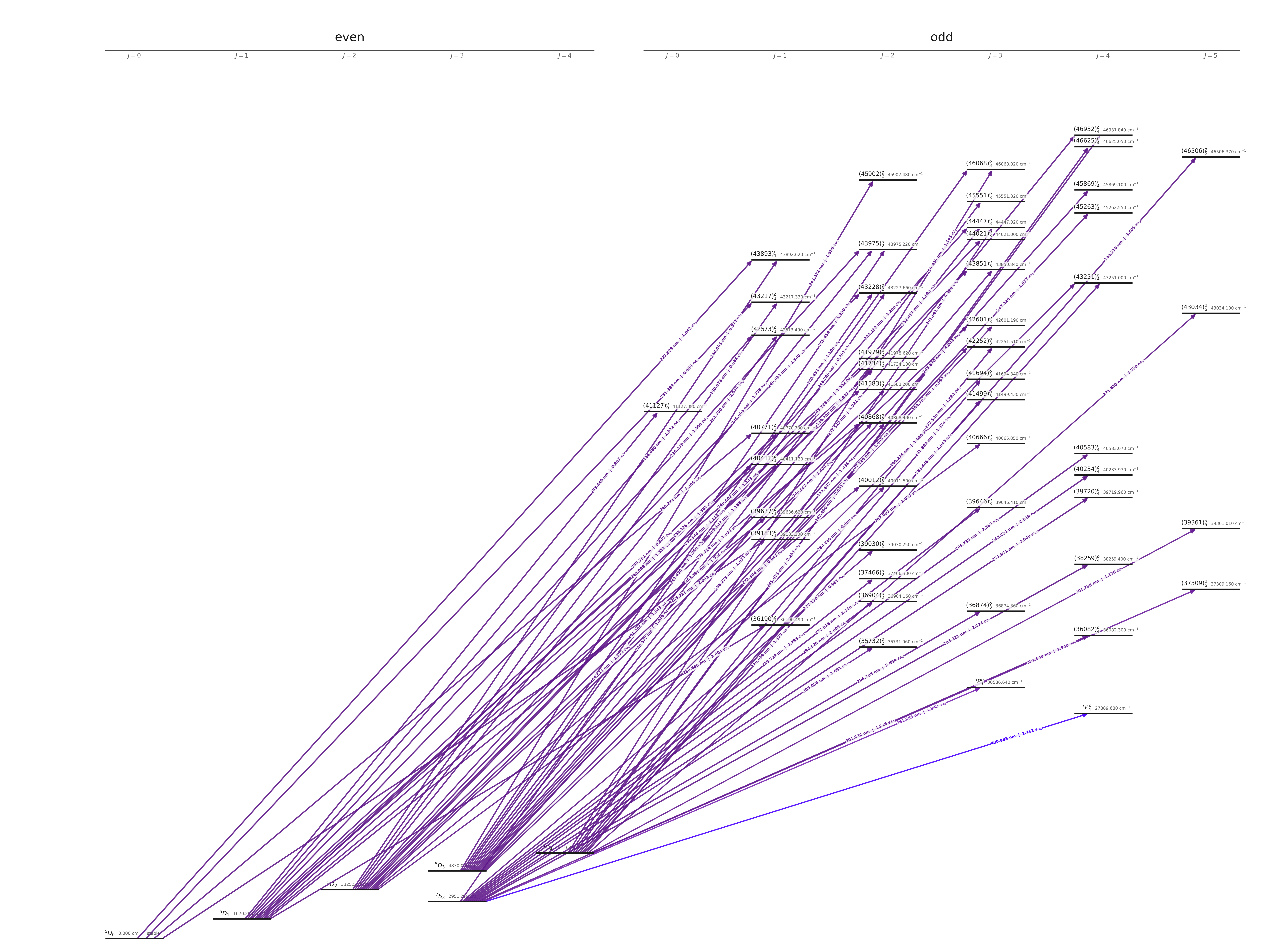


Tungsten

W (W I). Strongest classified transitions below 2 μm (NIST): 55 levels, 70 lines.

W⁺ ionisation limit 63427.70 cm⁻¹ = 7.86403 eV (157.660 nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	[U er U'] (ea0)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
⁵ D ₃ - (46068)½	242.4947	242.4211	1236.2847	1.960	7.800e+07	-	-	
⁵ D ₃ - (45902)½	243.4720	243.3982	1231.3220	1.956	1.074e+08	-	-	
⁵ D ₃ - (45869)½	243.6701	243.5962	1230.3213	4.043	2.544e+08	-	-	
⁵ D ₁ - (42573)½	244.4796	244.4056	1226.2471	1.372	8.700e+07	-	-	
⁵ D ₀ - (40771)½	245.2737	245.1994	1222.2772	1.305	7.800e+07	-	-	
⁵ D ₄ - (46932)½	245.6247	245.5504	1220.5303	2.237	7.600e+07	-	-	
⁵ D ₂ - (43975)½	246.0043	245.9299	1218.6471	1.778	8.600e+07	-	-	
⁵ D ₄ - (46625)½	247.4897	247.4149	1211.3330	2.831	1.190e+08	-	-	
⁵ D ₄ - (46506)½	248.2188	248.1438	1207.7751	3.505	1.480e+08	-	-	
⁵ D ₁ - (41127)½	253.4399	253.3637	1182.8938	0.887	9.800e+07	-	-	
⁵ D ₂ - (42573)½	254.7903	254.7138	1176.6242	2.070	1.750e+08	-	-	
⁵ D ₀ - (39183)½	255.2114	255.1349	1174.6828	2.093	1.780e+08	-	-	
⁷ S ₃ - (39646)½	272.5158	272.4351	1100.0920	2.710	1.050e+08	-	-	
⁷ S ₃ - (37466)½	289.7290	289.6441	1034.7340	2.793	1.300e+08	-	-	
⁷ S ₃ - (36904)½	294.5259	294.4398	1017.8814	2.609	1.080e+08	-	-	
⁷ S ₃ - (36874)½	294.7846	294.6985	1016.9881	2.694	8.200e+07	-	-	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Arrow label: vacuum wavelength | reduced dipole matrix element [U||er||U'] in ea0

Matrix elements are measured or NIST values: * = high-accuracy theory, = = model calculation.

Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

